Adaptive Sampling for Efficient Multi-Scale Network Centrality Analysis

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February 11, 2026

Abstract

Localised network centrality is routinely computed at multiple distance thresholds, from pedestrian to metropolitan scales. The analysis is particularly expensive at long distances, where each node's shortest-path traversal must explore a large reachable subgraph containing many thousands of nodes. Source-sampling — running traversals from a reduced fraction of nodes and rescaling the result — can yield large speedups through a statistical approximation of the full calculation. At long distances, the abundance of source contributions makes estimates reliable even from significantly smaller samples.

This raises two practical questions: when does a reachable subgraph contain enough nodes for statistical sampling to be sufficiently accurate, and what sampling probability should be used for a given distance once this threshold is reached? We propose a simple, per-distance schedule based on a single, interpretable quantity: reach r, the typical number of nodes within the distance threshold. A Hoeffding-style analysis suggests that the number of sampled sources needed for a given tolerance scales as $k = O(\log r/\varepsilon^2)$, which we convert into a sampling probability $p = \min(1,\, k/r)$.

On the Greater London (294,000 nodes) and Greater Madrid street networks at distances from 1–20 km, using $\varepsilon = 0.1$ yields strong rank agreement with the ground truth (Spearman $\rho \geq 0.97$) and speedups of up to 63.2×. The approach is implemented in the open-source `cityseer` Python package.

Keywords: network centrality, sampling, urban morphology, betweenness, harmonic closeness, computational efficiency

1 Introduction

Urban morphological analysis makes use of localised network centrality (typically variants of closeness or betweenness computed within distance thresholds) to reveal patterns of accessibility and flow in street networks. Because these metrics are typically evaluated at multiple scales, from pedestrian (∼500 m) to metropolitan (∼20 km), the cost is driven by the longest distances, where each node's traversal must explore a large reachable subgraph. For a network of 294,000 nodes, a single multi-scale centrality analysis can take hours.

Each source's traversal contributes to the centrality estimates of all nodes it reaches, so running traversals from a random fraction of sources and rescaling yields approximate estimates for the entire network. The typical number of reachable nodes — which we call *reach* r — varies with the distance threshold and governs both the cost and the effectiveness of this strategy. At short distances, reach is small, traversals are cheap, and there is little to gain from sampling. At long distances, reach is large and traversals are expensive, but each node's estimate also draws on contributions from many sources, making sampling both most beneficial and most reliable.

Source-sampling has been analysed for global centrality, but the localised setting offers a natural refinement: reach r varies with the distance threshold, so the required sample size can

be calibrated per distance. Building on the Eppstein–Wang analysis (Eppstein and Wang, 2004), we replace the global node-count n with reach r, obtaining a per-distance schedule with sample complexity $O(\log r/\varepsilon^2)$ (Section 3).

Our contributions are:

1. We derive a reach-based, per-distance source-sampling schedule for localised centrality: for each distance, compute $k = \log(2r/\delta)/(2\varepsilon^2)$ and $p = \min(1, k/r)$.
2. We show empirically that, at $\varepsilon = 0.1$, the resulting probabilities yield strong rank preservation for localised harmonic closeness and betweenness.
3. We validate on the Greater London (294,000 nodes) and Greater Madrid networks at distances from 1–20 km, observing speedups of up to 63.2× while maintaining $\rho \geq 0.97$.

The approach is implemented in the open-source `cityseer` Python package.

Paper structure. Section 2 defines the centrality measures and experimental protocol. Section 3 derives the reach-based sampling schedule, analyses its error structure on synthetic networks, and positions it relative to prior bounds. Section 4 reports validation on two large street networks, followed by practical guidance and limitations in Section 5.

2 Methods

We consider two centrality measures, both computed with a distance cutoff $d_{\max}$ as is common in urban network analysis.

Harmonic closeness (Rochat, 2009; Freeman, 1979) measures how near a node is to the rest of its distance-bounded neighbourhood: nodes with many nearby destinations receive higher values. It sums inverse distances to all nodes within the threshold:

$$C_H(v;\, d_{\max}) = \sum_{\substack{u \neq v \\ d(v,u) \leq d_{\max}}} \frac{1}{d(v,u)} \tag{1}$$

Betweenness (Freeman, 1977) measures how often a node lies on shortest routes between other nodes, capturing potential "through-movement" within the distance-bounded neighbourhood. It counts the fraction of shortest paths (with total length at most $d_{\max}$) that pass through each node:

$$C_B(v;\, d_{\max}) = \sum_{\substack{s \neq t,\ s \neq v,\ t \neq v \\ d(s,t) \leq d_{\max}}} \frac{\sigma_{st}(v)}{\sigma_{st}} \tag{2}$$

where σ_{st} is the number of shortest paths from s to t, and $\sigma_{st}(v)$ is the number passing through v. Common distance thresholds range from 400 m to 20 km.

A quantity central to the sampling analysis is *reach*: the mean number of nodes within $d_{\max}$ (including the source node itself),

$$r = \frac{1}{N} \sum_{v} |\{u : d(v,u) \leq d_{\max}\}| \tag{3}$$

In the Greater London network, reach ranges from 2,783 at 5 km to 43,165 at 20 km. This order-of-magnitude variation across distance thresholds directly governs the relationship between sample size and accuracy.

Synthetic networks. We generate synthetic networks using 3 topology templates: trellis (high connectivity, representing planned urban cores), tree (low connectivity, representing suburban sprawl), and linear (intermediate connectivity). These span the range of urban street

network structure. Each topology is tiled to create networks of approximately 12 km extent; we evaluate accuracy on a central 4 km×4 km window to reduce edge effects. We test 7 distance thresholds from 250 m to 4,000 m, and 12 sampling probabilities from 2.5% to 100%.

Boundary effects. Nodes near network edges have artificially lower centrality because their catchments are truncated. We address this by applying an inward buffer from the convex hull of all nodes, equal to the maximum analysis distance (4 km for synthetic networks, 20 km for validation). Only nodes inside this buffered zone are marked as "live" and included in accuracy calculations, ensuring all live nodes have complete catchments at all tested distances. This approach is consistent with common practice for localised network analysis (Gil, 2017).

Accuracy metrics. We measure sampling accuracy using two complementary metrics: Spearman rank correlation and normalised additive error.

Spearman rank correlation (ρ) between sampled and ground-truth centrality values:

$$\rho = 1 - \frac{6 \sum_i (R_i^{\text{sampled}} - R_i^{\text{true}})^2}{N(N^2 - 1)} \tag{4}$$

where R_i denotes the rank of node i. Spearman correlation is appropriate because many urban applications primarily use *relative* rankings (e.g., identifying the most central streets) rather than absolute values. Source-sampling also tends to estimate high-centrality nodes more precisely because they receive contributions from many sampled sources, which helps preserve the top of the ranking.

Normalised additive error $\varepsilon_{\text{obs}} = \max_v |C^{\text{sampled}}(v) - C^{\text{true}}(v)| \,/\, C_{\max}$, where $C_{\max}$ is an upper bound on the centrality value within the distance-bounded neighbourhood: $r(r-1)$ for betweenness and r for harmonic closeness. This normalised error lies in $[0, 1]$, is directly comparable to the ε parameter in the Hoeffding bound, and serves as the formal accuracy measure.

Validation networks. We validate the model on two real-world street networks. The primary validation uses the Greater London Authority (GLA) road network, sourced from Ordnance Survey Open Roads data. After preprocessing (removing filler nodes, consolidating nearby nodes, and removing dangles), the network comprises approximately 294,000 nodes. We apply a 20 km inward buffer to eliminate edge artifacts. As an external validation, we also test the model on the Greater Madrid metropolitan street network following the same protocol.

3 A Reach-Based Sampling Model

Figure 1 illustrates why sampling-based centrality is attractive. Panel A shows that accuracy (Spearman ρ) increases with sampling probability across all distance thresholds, but the curves flatten well before 100% sampling. At 4000 m distance, $\rho \geq 0.95$ is achieved at just 10% sampling, a 10× potential speedup. Panel B shows that the achievable speedup grows with analysis distance. The remainder of this section develops a sampling model inspired by the Eppstein–Wang bound that prescribes sampling probability as a function of reach alone.

A reach-based sampling schedule.

The Eppstein–Wang framework (Eppstein and Wang, 2004) provides the starting point. When k sources are sampled uniformly, each contributes a value in $[0, 1]$ (after normalisation) to each target node's centrality estimate. By Hoeffding's inequality (Hoeffding, 1963), the average of k such bounded contributions is concentrated around its expectation; a union bound extends this guarantee to all target nodes simultaneously. In the original global setting the union bound runs over all n nodes, yielding sample complexity $k = \log(2n/\delta)/(2\varepsilon^2)$.

For localised centrality with distance threshold $d_{\max}$, only nodes within a distance-bounded neighbourhood can receive non-zero contribution from a given sampled source. This motivates

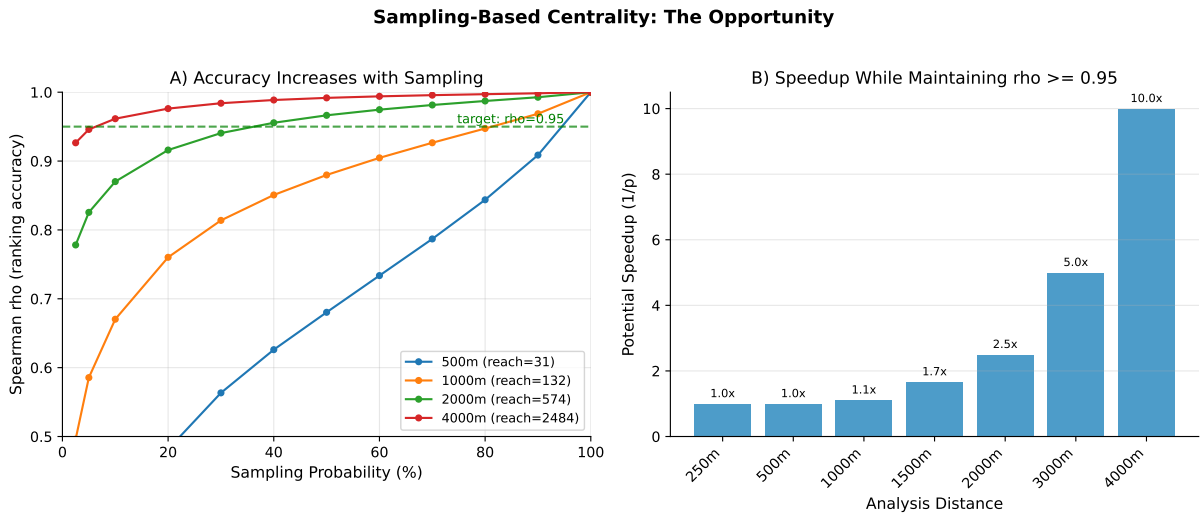

Figure 1: **The sampling opportunity.** (A) Accuracy vs. sampling probability for different analysis distances. The green dashed line marks the $\rho = 0.95$ target. (B) Potential speedup at each distance while maintaining $\rho \geq 0.95$.

replacing the global factor n with the mean reach r, giving

$$k = \frac{\log(2r/\delta)}{2\varepsilon^2} \tag{5}$$

with corresponding sampling probability

$$p = \min\left(1,\ \frac{k}{r}\right) \tag{6}$$

This model has no fitted parameters: ε (error tolerance) and δ (failure probability) are user-chosen conventions. At default values $\varepsilon = 0.1$, $\delta = 0.1$:

- **Low reach** ($r \lesssim 500$): $k > r$, so $p = 1$ and computation is exact (the schedule saturates naturally).
- **High reach**: p decreases as $O(\log r/r)$, so speedup $1/p$ grows as $O(r/\log r)$.

Figure 2 shows the model's prescribed effective sample size (k) and speedup ($1/p$) as functions of reach for three ε values. The logarithmic growth of k with r means that the sampling probability decreases rapidly with reach, producing speedups that grow with network size.

Table 1 provides the required effective sample sizes and sampling probabilities at three ε values across a range of reach values.

Error structure across reachability. Although the Hoeffding-style analysis targets a uniform additive error criterion, it is natural to ask how accuracy varies across the reachability distribution. We pool all nodes across all synthetic configurations at the Hoeffding-prescribed sampling probability ($\varepsilon = 0.1$) and bin them by absolute per-node reach. Figure 3 reveals a crossover: absolute error increases with reach, while normalised error decreases. This pattern has a natural interpretation. High-reach nodes sit at the centres of large local subgraphs; their centrality values are large sums over many source contributions, so absolute estimation errors accumulate. But these same nodes enjoy high effective sample sizes ($n_{\text{eff}} = p \cdot r$), so the errors are small relative to the centrality magnitude. Conversely, low-reach peripheral nodes have small absolute errors but larger relative errors. In practical terms, high-centrality arterials and junctions tend to receive higher precision, a property we term *precision scales with importance*.

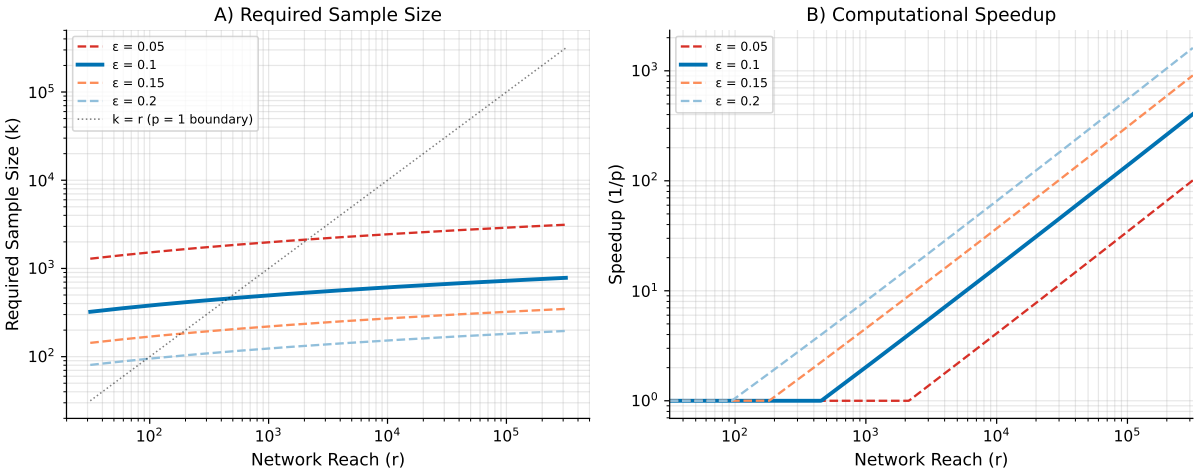

Figure 2: **The Hoeffding sampling model.** (A) Required effective sample size k vs. reach for three error tolerances. (B) Theoretical speedup $1/p$ vs. reach (log-log scale).

Table 1: Required effective sample sizes and sampling probabilities under the localised Eppstein–Wang bound at different additive error tolerances ($\delta = 0.1$).

Reach	**EW** ε=0.05	p	**EW** ε=0.1	p	**EW** ε=0.2	p
100	1,520	100.0%	380	100.0%	95	95.0%
500	1,842	100.0%	460	92.1%	115	23.0%
1,000	1,980	100.0%	495	49.5%	123	12.4%
3,000	2,200	73.3%	550	18.3%	137	4.6%
10,000	2,441	24.4%	610	6.1%	152	1.5%
30,000	2,660	8.9%	665	2.2%	166	0.6%

The Hoeffding model prescribes $k = \log(2r/\delta)/(2\varepsilon^2)$, $p = \min(1, k/r)$. At the default $\varepsilon = 0.1$, $\delta = 0.1$, the model has zero fitted parameters.

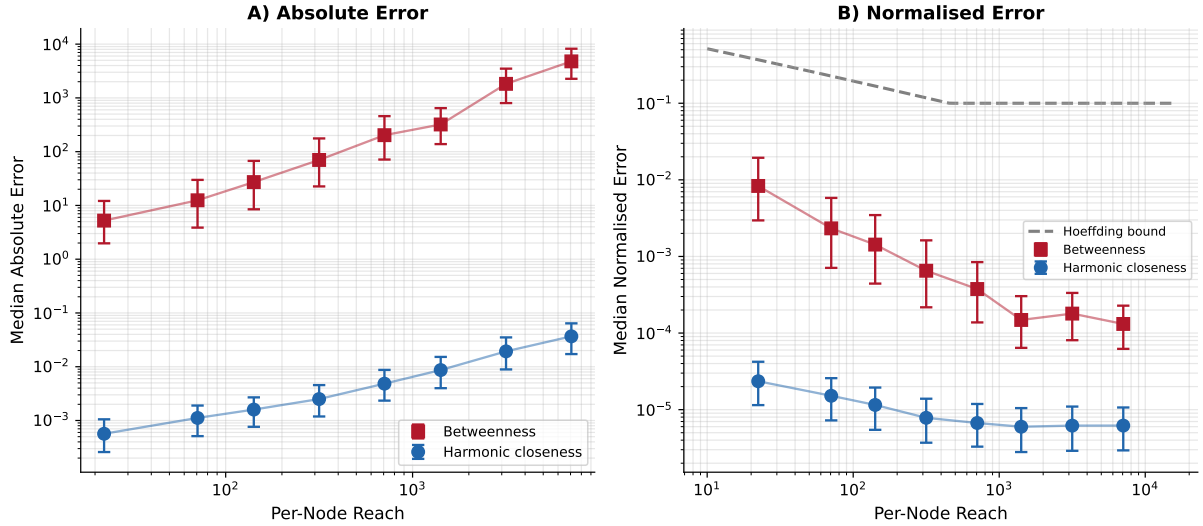

Figure 3: **Error crossover: precision scales with importance.** Per-node errors pooled across all synthetic configurations at the Hoeffding-prescribed sampling probability ($\varepsilon = 0.1$), binned by absolute per-node reach (log scale). Error bars show interquartile range across nodes within each bin. (A) Absolute error increases with reach: high-centrality nodes accumulate larger absolute errors because their centrality values are large sums over many source contributions. (B) Normalised error decreases with reach: relative error shrinks because high-reach nodes enjoy higher effective sample sizes ($n_{\text{eff}} = p \cdot r$). The dashed grey line shows the Hoeffding bound ($\varepsilon = \sqrt{\log(2r/\delta)/(2 \cdot n_{\text{eff}})}$, $\delta = 0.1$): observed median errors are two to three orders of magnitude below the worst-case prediction. Normalisers are r for closeness and $r(r-1)$ for betweenness.

This error structure helps explain why an additive-error schedule also preserves rank order. The nodes with appreciable normalised error tend to fall at intermediate reach, where centrality values are comparatively clustered and small rank swaps contribute little to Spearman ρ. Meanwhile, the high-reach nodes whose ranking matters most receive the highest precision, and low-reach peripheral nodes approach exact computation.

Relationship to prior bounds. Table 2 places our approach in the context of the existing sampling literature.

Table 2: Comparison of sampling approaches for centrality approximation. All prior methods target global centrality (no distance cutoff) on general graphs. Our approach adapts the Eppstein–Wang bound to localised centrality.

Method	Metric	Guarantee	Sample complexity	Key parameter
Eppstein & Wang (2004)	Closeness	Additive, all nodes	$O(\log n \,/\, \varepsilon^2)$	Node count n
Bader et al. (2007)	Betweenness	Relative, per vertex	$O(1 \,/\, \varepsilon^2 b(v))$	Betweenness $b(v)$
Riondato & Korn. (2016)	Betweenness	Additive, all nodes	$O(\log \mathrm{VD} \,/\, \varepsilon^2)$	Vertex diam. VD
KADABRA (2019)	Betweenness	Additive, all nodes	Adaptive	Diameter D
SILVAN (2023)	Betweenness	Non-uniform	Variance-dependent	Path variance
This work	**Both**	**Additive, all nodes**	$\boldsymbol{O(\log r \,/\, \varepsilon^2)}$	**Reach** r

All of these methods share a common pattern: sample a subset of computations, aggregate contributions at each node, and bound the resulting error. They differ in the unit of computation. Source-sampling methods (Eppstein–Wang, Bader et al., Brandes and Pich (Brandes and Pich, 2007)) run a full shortest-path tree from each sampled node, accumulating contributions at every

reached target; Cohen et al. (Cohen et al., 2014) apply the same source-to-target pattern for global closeness, combining it with pivot-based structural estimation. Path-sampling methods (Riondato and Kornaropoulos, 2016; Borassi and Natale, 2019; Pellegrina and Vandin, 2023) instead sample individual shortest paths between random node pairs, incrementing betweenness estimates along each path.

For localised centrality, source-sampling is the natural primitive: a single distance-bounded Dijkstra tree produces both closeness and betweenness contributions for all nodes within reach. Among the source-sampling bounds, Eppstein–Wang provides simultaneous additive guarantees at all nodes, while Bader et al. provide a tighter relative guarantee but only for a single target vertex. We adapt the Eppstein–Wang framework, replacing the global node count n with reach r to obtain a per-distance schedule.

Empirically, on the street networks studied here, the resulting schedule at $\varepsilon = 0.1$ is *practical*: observed errors are substantially smaller than the worst-case prediction (Figure 3B). This gap indicates that the Hoeffding-style analysis is conservative in these settings, making $\varepsilon = 0.1$ a pragmatic default when rank preservation is the primary objective.

Algorithm. Algorithm 1 summarises the complete procedure.

Algorithm 1 Adaptive Per-Distance Sampling for Network Centrality

Require: Network $G = (V, E)$, distance thresholds $D = \{d_1, \ldots, d_m\}$
Require: Error tolerance ε (default 0.1), confidence δ (default 0.1)
Ensure: Centrality estimates $\hat{C}[d]$ for each $d \in D$

```
Phase 1: Reach Estimation
for each distance d ∈ D do
    Compute reach r_d ← mean nodes within distance d
end for
Phase 2: Adaptive Sampling
for each distance d ∈ D do
    Compute required sources: k_d ← log(2r_d/δ)/(2ε^2)
    Compute sampling probability: p_d ← min(1, k_d/r_d)
    Sample sources S_d ⊆ V with probability p_d
    Compute centrality from S_d: Ĉ[d] ← CENTRALITY(G, S_d, d)
    Scale estimates: Ĉ[d] ← Ĉ[d]/p_d
end for
return {Ĉ[d] : d ∈ D}
```

The reach r_d is typically a byproduct of the centrality computation itself; when a quick estimate is needed beforehand, a pilot sample of ≈ 100 nodes distributed on a square kilometre grid was adequate for the networks in this study.

For example, on the Greater London network (Section 4), the model prescribes $p = 1.0$ at 1–2 km (reach $\lesssim 500$), $p \approx 19.6\%$ at 5 km (reach $\approx$ 2,783), and $p \approx 1.6\%$ at 20 km (reach $\approx$ 43,165).

4 Validation

The sampling schedule is calibrated to additive error ε, but urban applications typically care about rank order. We therefore report Spearman ρ as the primary validation metric: if the schedule controls additive error, and the error structure concentrates precision at high-centrality nodes (Section 3), then rank preservation should follow.

We evaluate on the GLA and Greater Madrid networks described in Section 2. For the GLA network, we compute ground-truth centrality ($p = 1.0$) and sampled centrality at the Hoeffding-prescribed probability ($\varepsilon = 0.1$, $\delta = 0.1$) at each distance threshold. For the Madrid network,

we repeat each configuration with five random seeds and report the mean ± standard deviation to assess stability.

Greater London. Table 3 presents validation results for the Greater London network at five distances from 1–20 km. At all distances, both harmonic closeness and betweenness achieve $\rho > 0.95$, with the minimum observed $\rho = 0.98$ (betweenness, 20 km). At short distances (1–2 km), the model prescribes $p = 1.0$ (exact computation). At longer distances, the prescribed sampling probabilities decrease with reach, and speedups grow from 5.1× at 5 km to 63.2× at 20 km.

Table 3: Hoeffding model validation on Greater London network ($\varepsilon = 0.1$, $\delta = 0.1$).

Distance	**Reach**	**Hoeff.** p	$\rho_{\mathbf{close}}$	$\rho_{\mathbf{betw}}$	**Speedup**	$\rho \geq 0.95$**?**
1km	100	100.0%	0.9948	0.9983	1.0×	✓
2km	411	100.0%	0.9979	0.9993	1.0×	✓
5km	2,783	19.6%	0.9875	0.9912	5.1×	✓
10km	11,590	5.3%	0.9930	0.9846	18.8×	✓
20km	43,165	1.6%	0.9959	0.9824	63.2×	✓

Network: Greater London Area, 294,486 nodes, 20km live node buffer. Hoeffding model: $k = \log(2r/\delta)/(2\varepsilon^2)$, $p = \min(1, k/r)$.

Betweenness is consistently the binding constraint: at every distance, $\rho_{\text{betw}} < \rho_{\text{close}}$. This is expected: betweenness contributions are sparser and heavier-tailed than closeness contributions. Most sources route zero shortest paths through a given node, while a few contribute heavily, resulting in higher per-source variance. The Hoeffding bound, which treats both metrics identically, is therefore more conservative for closeness.

Greater Madrid. To assess transfer beyond London, we applied the identical protocol to the Greater Madrid metropolitan street network. Table 4 presents the results. The model achieves $\rho > 0.95$ for both closeness and betweenness at all tested distances. The minimum observed $\rho = 0.9794$ (betweenness, 20 km). Speedups range from 2.8× at 5 km to 24.2× at 20 km, lower than GLA because the Madrid network is smaller (fewer reachable nodes at each distance).

Table 4: Hoeffding model validation on Greater Madrid network ($\varepsilon = 0.1$, $\delta = 0.1$).

Distance	**Reach**	**Hoeff.** p	$\rho_{\mathbf{close}}$	$\rho_{\mathbf{betw}}$	**Speedup**	$\rho \geq 0.95$**?**
5km	1,451	35.4%	0.9969	0.9949	2.8×	✓
10km	5,219	11.1%	0.9974	0.9872	9.0×	✓
20km	15,300	4.1%	0.9976	0.9794	24.2×	✓

Network: Greater Madrid, 99,166 nodes, 20km live node buffer. Hoeffding model: $k = \log(2r/\delta)/(2\varepsilon^2)$, $p = \min(1, k/r)$.

5 Discussion and Conclusion

Practical guidance. For users of the `cityseer` library, the sampling model is applied automatically. For manual implementations, follow Algorithm 1. Figure 4 provides visual lookup charts for common scenarios, and Table 5 gives a numerical reference for representative reach values.

As a simple heuristic: at reach below ∼500, $p = 1.0$ (exact computation). At reach 1,000–10,000, sample 5–50% (2–16× speedup). At reach above 10,000, sample below 6% (> 16×

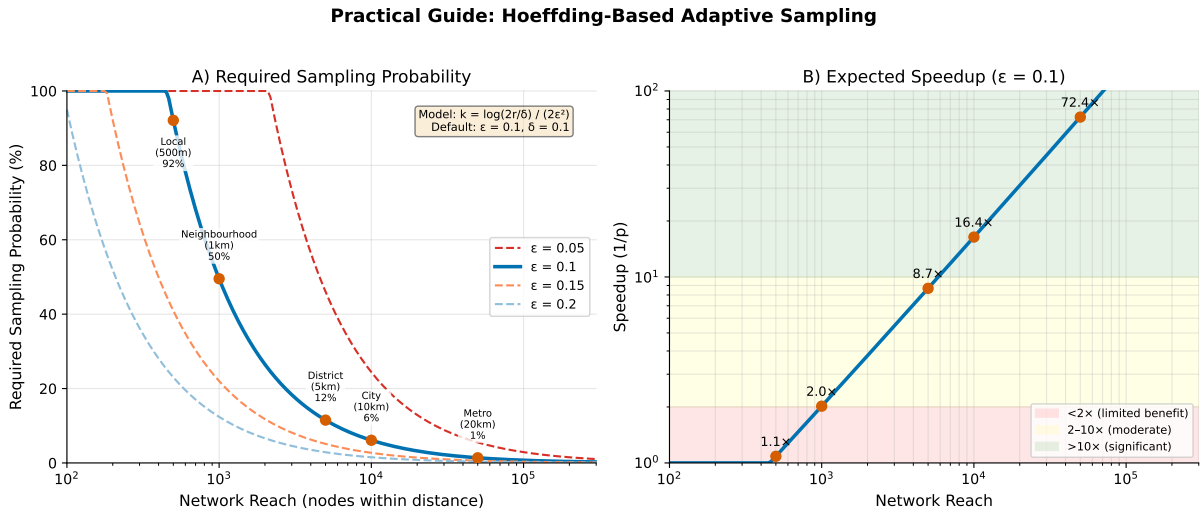

Figure 4: **Practical guidance.** (A) Required sampling probability by network reach, with common urban scenarios annotated. (B) Expected computational speedup.

Table 5: Practical sampling lookup table ($\varepsilon = 0.1$, $\delta = 0.1$).

Reach	k	p (%)	**Speedup**
100	380	100.0	1.0×
200	415	100.0	1.0×
500	461	92.1	1.1×
1,000	495	49.5	2.0×
2,000	530	26.5	3.8×
5,000	576	11.5	8.7×
10,000	610	6.1	16.4×
20,000	645	3.2	31.0×
50,000	691	1.4	72.4×
100,000	725	0.7	137.8×

Usage: Find your network reach (nodes within analysis distance), read off the required sampling probability.
Model: $k = \log(2r/\delta)/(2\varepsilon^2)$, $p = \min(1,\ k/r)$. At low reach, $k > r$ so $p = 1$ (exact computation).

speedup). Users requiring higher accuracy can reduce ε (e.g., $\varepsilon = 0.05$ roughly quadruples the required samples) or average 2–3 replications with different random seeds.

Sampling is not appropriate when exact centrality values (not approximations) are required, when centrality is needed for specific nodes rather than the full network, or when the network is small enough ($<$ 1,000 nodes) that full computation is already fast.

Limitations. Our approach rests on several assumptions. First, we assume uniform random sampling of source nodes. Non-uniform strategies (e.g., stratification by location or weighting toward high-centrality sources) could potentially achieve higher accuracy with fewer samples, but they would require proxies for node importance that are typically unavailable before computation.

Second, the schedule uses mean reach r as a pragmatic substitute in the logarithmic factor; more conservative choices could replace r with an upper quantile or maximum distance-bounded reach. Third, the model treats the network as statistically homogeneous: accuracy depends on global reach rather than local structure. Validation on the GLA and Madrid networks, which include dense urban cores and sparse suburbs, suggests the schedule is conservative overall, but accuracy may vary locally.

Fourth, our experiments used shortest-path distance; angular (simplest-path) distance, common in space syntax analysis, may exhibit different accuracy characteristics and warrants separate validation. Fifth, validation is limited to two European cities; networks with substantially different morphology (e.g., North American grids, informal settlements) would strengthen external validity. Finally, betweenness shows thinner margins than closeness at high reach: the minimum observed $\rho = 0.9794$ (Madrid betweenness, 20 km) remains above the 0.95 threshold but leaves less headroom than closeness.

Future work. Several extensions could build on this work. Adaptive per-node sampling, where sampling probability varies with local reach, could improve efficiency in heterogeneous networks. Importance sampling (weighting toward influential sources) could reduce variance. Online stopping criteria, as in KADABRA (Borassi and Natale, 2019), could replace a priori calibration with adaptive termination based on observed convergence; adapting such methods to localised, distance-bounded centrality would require new theoretical development. Systematic validation for angular distance, and extension to other centrality measures (eigenvector centrality, PageRank), would broaden applicability. Finally, variance-dependent analyses such as SILVAN (Pellegrina and Vandin, 2023), which replace worst-case combinatorial quantities with the observed estimator variance, are a promising direction for tightening localised bounds on well-structured network classes.

A simple reach-based source-sampling schedule yields strong rank preservation ($\rho \geq 0.97$) for localised centrality on the street networks studied here, at $\varepsilon = 0.1$ and without fitted parameters. The error structure favours high-centrality nodes: high-reach nodes receive higher effective sample sizes and correspondingly more precise estimates, while low-reach peripheral nodes approach exact computation. Across the Greater London (294,000 nodes) and Greater Madrid networks at 1–20 km, speedups reach up to 63.2×, with all tested configurations exceeding $\rho \geq 0.95$. For city-scale networks where multi-scale centrality analysis previously required hours, this makes exploratory workflows and finer parameter sweeps practical.

The model is implemented in the open-source `cityseer` Python library.[1] Analysis scripts and validation data are available in the repository's `analysis/sampling/` directory.

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A Technical Details

Synthetic network generation. Synthetic networks are generated using tile-based templates. Each template defines a connectivity pattern within a single tile; tiles are then arrayed to create networks of the desired extent.

Trellis A regular grid with diagonal connections, representing highly connected urban cores. Average degree ≈ 4.

Tree A hierarchical branching structure where each node connects to at most 3 others. Represents suburban cul-de-sac development. Average degree ≈ 2.3.

Linear A constrained grid with fewer cross-connections than trellis. Intermediate connectivity. Average degree ≈ 3.

Networks are tiled 24×24 to achieve approximately 12 km extent, then the largest connected component is extracted. Node positions are perturbed slightly to avoid perfectly regular spacing.

GLA network preprocessing. The Greater London Authority road network is sourced from Ordnance Survey Open Roads data (GeoPackage format).

Preprocessing steps:

1. Remove invalid and empty geometries
2. Explode multi-part geometries
3. Build NetworkX graph from edges
4. Remove filler nodes (degree-2 nodes that don't represent junctions)
5. Remove dangling nodes (dead-end segments < 20 m)
6. Consolidate nearby nodes (threshold 1 m)
7. Extract largest connected component

Final network: $\approx$ 294,000 nodes.

Correctness verification against NetworkX. To ensure the correctness of our Rust-based centrality implementation in `cityseer`, we conducted systematic verification against the widely-used NetworkX library (Hagberg et al., 2008). For each synthetic network topology and distance threshold combination, we computed both harmonic closeness and betweenness centrality using our Rust implementation (via `cityseer` Python bindings) and NetworkX's reference functions.

For each node v in the test networks, we computed the absolute difference

$$\Delta_v = |C_{\text{cityseer}}(v) - C_{\text{NetworkX}}(v)| \tag{7}$$

The maximum difference across all nodes and all test configurations was $\max_v \Delta_v < 10^{-6}$, attributable to floating-point arithmetic precision.

Verification was performed across all three synthetic topologies, distance thresholds from 200 m to 3200 m, both shortest-path and angular distance heuristics, and both harmonic closeness and betweenness centrality.

Horvitz–Thompson estimator. When estimating population totals from samples, two classical approaches are the Horvitz–Thompson (H–T) estimator and the Hájek ratio estimator. Under Bernoulli (independent) source sampling with inclusion probabilities π_i, the Horvitz–Thompson estimator (Horvitz and Thompson, 1952) estimates a population total as $\hat{T}_{HT} = \sum_{i \in S} y_i/\pi_i$, where S is the set of sampled sources. This inverse-probability weighting yields an unbiased estimator whenever $\pi_i > 0$ for all sampled units.

The Hájek estimator (Hájek, 1971) instead uses a ratio form,

$$\hat{T}_H = N \cdot \frac{\sum_{i \in S} y_i/\pi_i}{\sum_{i \in S} 1/\pi_i}, \tag{8}$$

which is often lower-variance but can be slightly biased in finite samples (Cochran, 1977; Särndal et al., 2003). In `cityseer`, we use Horvitz–Thompson-style inverse-probability weighting because it matches the paper's $1/p$ rescaling description and extends directly to unequal inclusion probabilities (e.g., when combining a base sampling probability with per-source importance weights).

Implementation in `cityseer`. The sampling model is implemented in the `cityseer` Python library (version 4.23+). The adaptive function automatically probes reachability, computes sampling probabilities from the Hoeffding model, and scales results:

```
from cityseer.metrics import networks

gdf = networks.node_centrality_shortest_adaptive(
    network_structure,
    nodes_gdf,
    distances=[500,1000,2000,5000,10000],
    compute_closeness=True,
    compute_betweenness=True,
    epsilon=0.1,
    delta=0.1
)
```

For manual control, a fixed sampling probability can be specified by setting `sample_probability` in `node_centrality_shortest`. Helper functions for custom workflows are available in `cityseer.config`, including:

- `compute_hoeffding_p`
- `compute_sample_probs`
- `probe_reachability`
- `spatial_sample`

See the library documentation for details.

Reproducibility. All analysis scripts and validation results are available at analysis/sampling/. The script pipeline runs sequentially from `00_generate_cache.py` through `07_hoeffding_model_figure.py`, with each script's outputs documented in its header.